

GROUP PROJECT: ACTION PLAN

Personal Data Ecosystem for Learning Analytics
Introduction to Data Engineering — PR10

PART 1: IMPROVEMENT PLAN

WHAT WE HAVE (Already Delivered)

Report section 1.docx — Dataset Analysis
Report section 2.docx — Database Technology Comparison
Report section 3 (Final Version).docx — System Design
Report section 4 ETL Implementation.docx — ETL Code Documentation
Report section 5 (Test Report).docx — Test Results
Final_Report_Research_Findings.docx — Research Conclusions
Code/ folder: schema.sql, load_data.py, queries.py, load_log.txt
Data/ folder: 11 CSVs across 3 students + academic_performance.csv
README.md — Project overview and structure

WHAT IS MISSING (Per Rubric Feedback)

PROJECT_PLAN.md — Fixes PI 2.1 ("Plan to act is missing")

Docker Setup — Fixes PI 6.2 ("Infrastructure missing")

Dockerfile, docker-compose.yml, .env file

5-PHASE IMPROVEMENT ACTION PLAN

PHASE 1: PROJECT_PLAN.md

Sections to include:

Project Scope: Personal Data Ecosystem + Learning Analytics (89 days, 3 students, 1,371 rows)
Objectives: 5 Sections + ETL Pipeline + Docker + 5 Analytical Queries
Team Roles: Who worked on what (data collection, database, code, documentation)
Timeline: When each section/phase was completed (Sep 2024 - Jul 2026)
Milestones: Data collection (Week 1-6) → Section 1 → Section 2 → ... → Final Report

PHASE 2: DOCKER INFRASTRUCTURE

Create 3 files so anyone can run the project in 1 command

File 1: Dockerfile

```
FROM postgres:15 COPY Code/schema.sql /docker-entrypoint-initdb.d/ COPY Data/ /data/  
COPY Code/*.py /scripts/
```

File 2: docker-compose.yml

```
version: "3.8" services: postgres: image: postgres:15 environment: POSTGRES_DB:  
learning_analytics POSTGRES_USER: postgres POSTGRES_PASSWORD: (from .env)  
ports: - "5432:5432" volumes: - postgres_data:/var/lib/postgresql/data
```

File 3: .env (template)

```
DB_HOST=localhost DB_PORT=5432 DB_NAME=learning_analytics DB_USER=postgres  
DB_PASSWORD=your_password
```

Feedback says "infrastructure missing" — Docker allows reproducibility

PHASE 3: UPDATE README.md

Add "Quick Start with Docker" section:

```
docker-compose up -d docker exec -it (container_id) psql -U postgres -d learning_analytics -f  
/docker-entrypoint-initdb.d/schema.sql python /scripts/load_data.py python /scripts/queries.py
```

PART 2: FROM SCRATCH PLAN

WEEK 1-2: PLANNING & ENVIRONMENT SETUP

Week 1 Step 1: Create Project Plan

Define scope: Personal Data Ecosystem for Learning Analytics
Define timeline: 89 days (Sep 2 - Nov 29, 2024)
Assign roles: Who collects WHOOP? MyFitnessPal? Academic data?
Define deliverables: 5 Sections + Code + Docker + Final Report

Week 1 Step 2: Setup Technical Environment

Create GitHub repo: maestrosifu/Data-Engineering-Project
Create folder structure: /Code, /Data, /Reports, /Docker
Install PostgreSQL 15 locally
Install Python: pandas, psycpg2

WEEK 2: DATA COLLECTION (89 days continuous)

Export WHOOP wearable data weekly: 3 files per student (cycles, sleep, workouts)
Export MyFitnessPal nutrition logs: 3 files (meal logs per student)
Collect academic performance: grades from 5 courses, 3 exams = 45 records
Push CSVs to GitHub /Data folder Result: 1,371 total rows

WEEK 2: SECTION 1 DATASET ANALYSIS

Analyze 11 CSV files: data types, granularity, missing values, statistics
Document per-student patterns: sleep (7.85h vs 5.42h), recovery (78% vs 32%)
Create Report section 1.docx

WEEK 2: SECTION 2 DATABASE TECHNOLOGY COMPARISON

Research 4 databases: PostgreSQL, InfluxDB, MongoDB, Cassandra
Compare on: join support, ACID, schema flexibility, time-series optimization
Select PostgreSQL: multi-table joins needed for Learning Analytics
Create Report section 2.docx

WEEK 3: SECTION 3 SYSTEM DESIGN

Design 6-table schema: students, whoop_cycles, whoop_sleep, whoop_workouts, nutrition_logs, academic_performance
Document GDPR compliance: cascading deletes, pseudonymization (S001/S002/S003)
Document ISO 27001: encryption, access control, backup strategy, logging
Document distributed architecture: single primary, future replication standby
Create Report section 3 (Final Version).docx

WEEK 3: SECTION 4 ETL IMPLEMENTATION

Code schema.sql: 6 tables, constraints (CHECK, NOT NULL), indexes, cascades
Code load_data.py: read 11 CSVs, type converters (to_float_or_none, to_bool), idempotency (ON CONFLICT)
Code queries.py: 5 analytical queries with JOINS and aggregations
Execute and verify: load_log.txt shows 1,343 rows, 0 errors
Create Report section 4 ETL Implementation.docx

WEEK 4: SECTION 5 TEST REPORT

Test 1: Run load_data.py and verify all 1,343 rows loaded
Test 2: Run load_data.py again, confirm no duplicates (idempotency)
Test 3: Run all 5 queries and measure execution times (< 10 ms each)
Test 4: Verify GDPR delete (cascade across all 6 tables)
Create Report section 5 (Test Report).docx

WEEK 5: FINAL REPORT & RESEARCH FINDINGS

Extract findings from 5 queries: sleep, activity, nutrition patterns
Answer main question: How do lifestyle habits relate to academic performance?
Document per-student conclusions: S001 (avg 8.36 grade) vs S003 (5.51)
Create Final_Report_Research_Findings.docx

WEEK 6: INFRASTRUCTURE & DOCUMENTATION

Create Dockerfile: PostgreSQL 15 base image, schema.sql, CSVs, Python scripts
Create docker-compose.yml: PostgreSQL service, volumes, port 5432
Create .env template: DB_HOST, DB_PORT, DB_NAME, DB_USER, DB_PASSWORD
Create PROJECT_PLAN.md: scope, objectives, timeline, roles, milestones
Update README.md with Quick Start (Docker commands)

WEEK 7: PRESENTATION & SUBMISSION

Create PPTX: 6 slides (We did the presentation before to give information about how the project was going, but we don't remember which week was it)
Design poster (readability, information hierarchy)
Push everything to GitHub and submit